MAT627 Chapter 2.7 Programming Assignment

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April 20, 2026

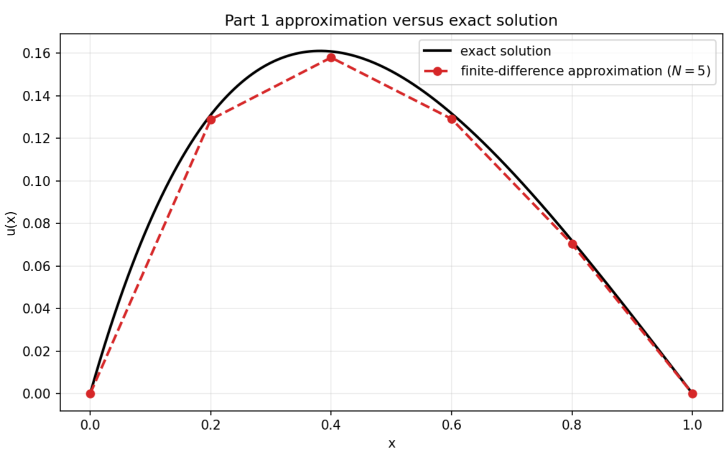

Figure 1: Part 1: nodal finite-difference approximation with $N = 5$ versus the exact solution.

1 Part 1

Table 1: Finite-difference solution for $-u'' + u = 4e^{-x} - 4xe^{-x}$ with $N = 5$.

x_k	U_k	$u(x_k)$	$u(x_k) - U_k$
0.2	1.288292×10^{-1}	1.309969×10^{-1}	2.167680×10^{-3}
0.4	1.580141×10^{-1}	1.608768×10^{-1}	2.862697×10^{-3}
0.6	1.291688×10^{-1}	1.317148×10^{-1}	2.545964×10^{-3}
0.8	7.036635×10^{-2}	7.189263×10^{-2}	1.526284×10^{-3}

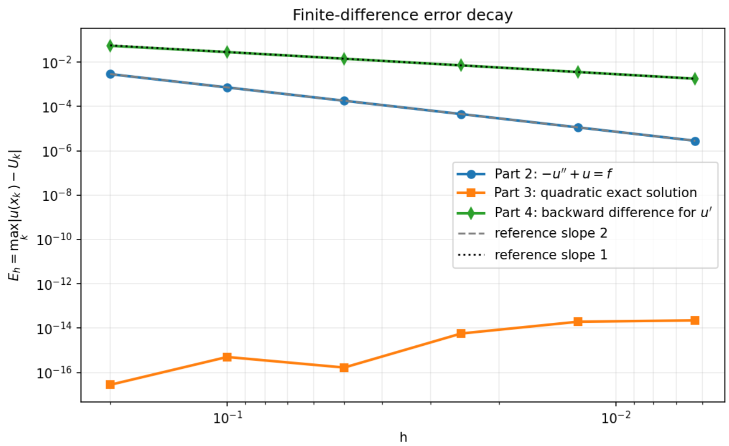

Figure 2: Error decay for the convergence studies in Parts 2–4.

2 Part 2

Discrete error for $-u'' + u = 4e^{-x} - 4xe^{-x}$ with exact solution $u(x) = x(1-x)e^{-x}$.

Table 2: Maximum nodal error for part 2.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	2.862697×10^{-3}	–
10	0.1	7.199515×10^{-4}	1.991403
20	0.05	1.803450×10^{-4}	1.997140
40	0.025	4.520950×10^{-5}	1.996062
80	0.0125	1.130343×10^{-5}	1.999865
160	0.00625	2.825924×10^{-6}	1.999966

3 Part 3

Discrete error for $-u'' + u = 2 + x - x^2$ with exact solution $u(x) = x(1-x)$.

Table 3: Maximum nodal error for part 3.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	2.775558×10^{-17}	–
10	0.1	4.996004×10^{-16}	–
20	0.05	1.665335×10^{-16}	–
40	0.025	5.717649×10^{-15}	–
80	0.0125	1.945666×10^{-14}	–
160	0.00625	2.231548×10^{-14}	–

4 Part 4

Discrete error for $-u'' + u' + u = -(x-1)(x^2-11)$ using the backward difference $(u(x) - u(x-h))/h$ for u'.

Table 4: Maximum nodal error for part 4.

N	h	$E_h = \max_k \lvert u(x_k) - U_k \rvert$	Rate
5	0.2	5.445658×10^{-2}	–
10	0.1	2.828373×10^{-2}	0.945134
20	0.05	1.418957×10^{-2}	0.995142
40	0.025	7.106748×10^{-3}	0.997569
80	0.0125	3.558012×10^{-3}	0.998118
160	0.00625	1.779658×10^{-3}	0.999471